

Master Learning Plan v10 FINAL

Medical-Imaging + Geometric/Molecular-ML — Research-Engineer Readiness — Master Learning Plan (v10 · FINAL)

Owner: MD Hashim · **Window:** mid-Aug 2026 → end-Dec 2027 · **Timeline (committed):** Block 1 complete ~**Apr 2027** (Tier-2 interview-ready) · Block 2 complete ~**Jul-Aug 2027** (Tier-3 loops) · Phase Q + everything complete ~**end-2027** · **Total effort:** ~2,530 focused hours (phases ~1,980 + threads ~550, run concurrently) · **Pace:** aggressive working weeks (~38-42 hrs/wk on top of the job) with **consolidation weeks that let the timeline stretch, never compress** · **Structure:** a job-critical spine (Blocks 1-2) + a **Quantum/QML track (Phase Q) run parallel / after Tier-2** + three robust continuous threads + the **Confusion Buffer** scaffold + an application overlay that starts from **month ~2**.

This is the definitive version. It locks every decision from the iterative review: **Research Engineer** as the near-term target (RS is PhD-gated, long-term); **no section below its v6 hours** (CV-expert and quantum restored in full, with added resources); the **geometric/molecular spike protected and deepened**; **Phase Q run parallel / after you reach Tier-2** so full QML depth never delays your ladder; robust **threads** so no interview surface feels shaky; the **Confusion Buffer** as your shock absorber for hard terminology; and a **dated Master Calendar** that lands Block 1 by April 2027 and everything by end-2027 while leaning on consolidation weeks. All v6/v7/v8/v9 URLs are preserved.

Changelog (v9 → v10 · FINAL)

Area	v9	v10 (final)
Phase Q sequencing	Two options (A integrated / B parallel)	Locked to parallel / after-Tier-2 — starts ~Apr 2027, runs alongside Block 2 + a concentrated block, done ~end-2027
Video / Tier-1 ladder	Video in Phase 8 with a "pull early" option	Pulled into Block 1 as Phase 2V (after segmentation) → clean Tier 0 → 1 → 2 ladder inside Block 1. Phase 8 keeps the rest of CV-expert (classical/geometric/3D/edge). No hours cut
Timeline	v6-aligned targets	Committed dated Master Calendar — Block 1 ~Apr 2027, Block 2 ~Jul-Aug 2027, all ~end-2027, with consolidation weeks scheduled
Pace philosophy	Sustainable, extend not compress	Same, made explicit — aggressive working weeks + consolidation weeks; the timeline stretches rather than the standards dropping
Everything else (v9)	Hours floor, added resources, robust threads, Confusion Buffer, RE-target, spike, QML branch, PhD/pub/risk	All preserved

MASTER CALENDAR (committed) — what happens when

Dates assume a mid-Aug 2026 start at an aggressive pace (~38-42 hrs/wk on working weeks) with a **consolidation week roughly every 5th week** (no new material — only re-derivation, glossary review, and re-implementation of shaky concepts). The **self-tests and Feynman gates are the true gates**: if a gate isn't passed, the timeline stretches — you do not advance on shaky foundations. Threads T/M/R run continuously from the Phase-1 self-test (~mid-Oct 2026). Interviewing is incremental from month ~2 and intensive in the back half.

Period	Primary focus	Milestone / tier
Aug 2026 (from ~18th)	Phase 0 — fundamentals rebuild + start the living glossary	—
~Early Sep 2026	Phase 0 self-test passed	► START APPLYING — Tier 0 (India medical-imaging: Qure.ai, SigTuple, GE, NVIDIA) on your existing applied skill
Sep - late Oct 2026	Phase 1 — math + NN from first principles (intuition-first) · [consolidation wk ~late Sep]	Phase-1 self-test → Threads T, M, R begin (~mid-Oct)
Late Oct - early Dec 2026	Phase 2 — medical image segmentation + Capstone 1 · [consolidation wk ~mid-Nov]	Capstone 1 done → Tier 0/1 interview-ready
Dec 2026	Phase 2V — imaging & video extension + Capstone 2 · [consolidation wk ~late Dec]	Capstone 2 done → Tier 1 unlocked (Recursion/phenomics; broader imaging)
Jan 2027	Phase 3 — DL theory & modern architectures · [consolidation wk ~late Jan]	Tier 1 (broad DL/CV) strengthened
Feb 2027	Phase 5 — production AI systems · [consolidation wk ~mid-Feb]	—
Mar - ~mid-Apr 2027	Phase 6 — geometric/molecular spike + Capstone 3 (flagship) · [consolidation wk ~mid-Mar]	Capstone 3 done → BLOCK 1 COMPLETE → Tier 2 RE interview-ready (Isomorphic / DeepMind-science)
~Mid-Apr 2027	Begin serious Tier-2 applications ; Phase Q begins (parallel-lite)	► Tier 2 loops start
Apr - May 2027	Phase 7 — GenAI & agents + Capstone 4 · Phase Q parallel-lite	Tier 2 broadened
May - Jun 2027	Phase 8 — CV-expert breadth (classical/geometric/3D/edge) + Capstone 5 · Phase Q continues · [consolidation wk ~mid-Jun]	Full CV depth locked

2027	Capstone 3 — Phase Q continues — consolidation wk — imaging	
Jun - ~Aug 2027	Phase 9 — interview crescendo (DSA, implement-from-scratch, ML system design, job-talk)	Phase-9 self-test → BLOCK 2 COMPLETE → Tier 3 top-lab loops. Intensive interviewing
Aug - Nov 2027	Phase Q concentrated block — deep QML build + Capstone Q — continue interviewing all tiers	QML career branch opens (QpiAI, IBM Quantum)
~Nov - Dec 2027	Phase Q complete; expertise grinding; closing offers	EVERYTHING COMPLETE — confident across Tiers 0-3 + QML branch

If you fall behind a gate: let the block stretch (you've chosen mastery over speed) — the back half (Block 2 + Phase Q, ~Apr→Dec) has slack because threads have matured and the load lightens. Better to hit Block 1 a few weeks late and *solid* than on-time and shaky.

The program at a glance

Block 1 — Foundations, domain, imaging-video & the spike (~1,135 hrs; done ~Apr 2027). Fundamentals rebuild → full intuition-first math → medical-imaging domain (+ Capstone 1) → imaging-video extension (+ Capstone 2) → DL theory → production systems → the **geometric/molecular spike** (+ Capstone 3, flagship). This block takes you Tier 0 → 1 → 2.

Block 2 — CV-expert breadth, GenAI & interview crescendo (~510 hrs + interviewing; done ~Jul-Aug 2027). GenAI & agents (+ Capstone 4) → full CV-expert breadth: classical/geometric/3D/edge (+ Capstone 5) → the research + big-tech interview crescendo. This block takes you to Tier 3.

Phase Q — Quantum/QML (~335 hrs; run parallel from ~Apr, concentrated ~Aug-Nov; done ~end-2027). A complete, intuition-first quantum + QML build from near-zero — full depth, off the job-critical path so it never delays your ladder — seeding a QML career branch (QpiAI) and deepening the spike's quantum-informed story.

Threads — robust (~550 hrs, from the Phase-1 self-test). DSA + C++ (~5 hrs/wk), implement-by-hand ML (~3 hrs/wk — highest-ROI + concept-cement), research output & networking (~2 hrs/wk). Sized so every interview surface feels fluent.

The Confusion Buffer. Your shock absorber for confusing terminology — intuition-first ordering, a living glossary, spaced re-derivation, a Feynman gate, an unblock ritual, consolidation weeks, understanding-gated checkpoints, and multi-teacher triangulation for the hardest topics.

Why this works — full depth, early value, sustainable

You named the core tension: a **vanilla model config hits a ceiling**, and you can't push past baseline — or explain *why* a model behaves as it does — without solid math and fundamentals. v10 gives the foundations the generous, intuition-first time they need for you, and full breadth everywhere so nothing feels shaky, **while** preserving early value: you apply from ~month 2 on your existing applied skill, and each capstone unlocks a higher tier (C1 → Tier 0/1; C2 → Tier 1; C3 → Tier 2; the crescendo → Tier 3). Phase Q stays off the job-critical path so full QML depth never delays the ladder. And the pace is sustainable-aggressive: hard working weeks buffered by consolidation weeks, with the timeline stretching rather than the standards dropping.

The Confusion Buffer — your shock absorber for hard concepts

Apply to every topic that feels slippery. It turns "I keep forgetting the terminology / I'm not intuitive about this" into a managed system. Use it constantly.

- Intuition-first ordering (never start rigorous cold).** (a) visual explainer first (3B1B, StatQuest, a good blog) → (b) rigorous source (MIT/Strang, a paper) → (c) implement/derive by hand. Starting rigorous before you have the picture is the #1 cause of slow-learner grind.
- Living glossary ("terminology war chest").** One doc. Every confusing term: one-line plain-English definition + tiny example + where it shows up. Add every session. Fixes "forget most of the terminology."
- Spaced re-derivation (Anki-style, for concepts).** Anything hard re-explained/re-derived at **1 day → 3 days → 1 week → 1 month**. Anki <https://apps.ankiweb.net/> or a dated checklist.
- The Feynman gate (the real "done" test).** Not done until you can explain it out loud, plain language, **to a 12-year-old, no notes**. If you can't, you've found the exact gap. (This *is* your interview-articulation training.)
- The unblock ritual (no grinding).** Stuck >30 min: log the exact confusion, then **switch teacher/modality** rather than re-reading. A different angle usually unlocks it fast.
- Consolidation weeks (scheduled ~every 5th week).** No new material — only re-derive, review the glossary, re-implement shaky concepts. Bank these; they keep confusion from compounding.
- Understanding-gated checkpoints.** Self-tests are pass/fail on *explain + derive + implement unaided*. **Don't advance until you pass** — the timeline stretches instead.
- Multi-teacher triangulation for the hardest ~6 topics** (2-3 teachers each; if one doesn't click, another will): Backprop — 3B1B NN https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi · micrograd <https://github.com/karpathy/micrograd> · d2l.ai <https://d2l.ai/> · Attention/vdK — Annotated Transformer <http://nlp.seas.harvard.edu/annotated-transformer/> · 3B1B <https://www.youtube.com/@3blue1brown> · Lilian Weng <https://lilianweng.github.io/> · SVD/eigen — 3B1B LA https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFItgF8hE_ab · MIT 18.06 <https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/> · MIT 18.065 <https://ocw.mit.edu/courses/18-065-matrix-met-hods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/> · Diffusion — Yang Song <https://yang-song.net/blog/2021/score/> · Lilian Weng <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/> · CS236 <https://www.youtube.com/playlist?list=PLoROMvovd4rPOWA-omMM6STXaWW4FvJT8> · Equivariance — Bronstein <https://geometricdeeplearning.com/lectures/> · e3nn tutorial https://blondegeek.github.io/e3nn_tutorial/ · Veličković https://www.youtube.com/results?search_query=petar+velickovic+gnn+lectures · VQE/barren plateaus — PennyLane demos https://pennylane.ai/qml/demos_quantum-chemistry · Quantum Country <https://quantum.country/qcvc> · Schuld https://www.youtube.com/results?search_query=maria+schuld+taking+stock+quantum+machine+learning

Threads (robust muscle-memory + output scaffold)

Start at the **Phase-1 self-test** (~mid-Oct 2026); run continuously to the end. Do them first thing, before the day's phase work. Flex to ~2-3 hrs combined only on consolidation/heavy weeks.

Thread T — DSA + C++ · ~5 hrs/week · so coding rounds never feel shaky (~275 hrs)

Stage T1 — C++ fundamentals (~40 hrs): types, references/pointers, RAII, STL, classes, templates, move semantics, smart pointers. [learncpp.com](https://www.learncpp.com/) (1-17) <https://www.learncpp.com/> · The Cherno <https://www.youtube.com/@TheCherno> · cppreference <https://en.cppreference.com/> · *Effective Modern C++* (Meyers) · CppCon <https://www.youtube.com/@CppCon> **Stage T2 — DSA in C++, sustained (~230+ hrs):** ~4-6 problems/wk, spaced → timed. NeetCode 150 <https://neetcode.io/> · LeetCode <https://leetcode.com/> · MIT 6.006 <https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/> · Abdul Bari https://www.youtube.com/@abdul_bari · CLRS · DesignGurus <https://www.designgurus.io/> · Striver A2Z + SDE sheet <https://takeuforward.org/> · Sean Prashad patterns <https://seanprashad.com/leetcode-patterns/> · CP Handbook <https://cses.fi/book/book.pdf> · CSes <https://cses.fi/problemset/> **Checkpoint:** core structures + standard patterns from memory in C++; a random LeetCode medium in ~25-30 min, hard in ~40-45.

Thread M — Implement-by-hand ML · ~3 hrs/week · highest-ROI + concept-cementor (~165 hrs)

Validated: DeepMind's RE coding round asks you to implement a loss (e.g. Focal Loss) from a blank file — your own domain loss. From a **blank file, no autocomplete, no AI**, one primitive in ~30-45 min, revisit spaced; add a weekly "build a small system" slot later. Rotating: backprop MLP · scaled-dot-product + multi-head attention · Dice/Focal/Tversky · cross-entropy from logits · Layer/BatchNorm · stable log-softmax · a sampler · a DDPM step · logistic/linear regression + gradients · k-means · PCA via SVD · SGD/Adam · positional encodings (sinusoidal/RoPE) · KV-cache · a message-passing GNN layer · a small VQE circuit · translate to JAX. d2l.ai · [labml.ai](https://nn.labml.ai/) <https://nn.labml.ai/> · Annotated Transformer <http://nlp.seas.harvard.edu/annotated-transformer/> · micrograd <https://github.com/karpathy/micrograd> · nanoGPT <https://github.com/karpathy/nanoGPT> · target rounds <https://igotanoffer.com/en/advice/google-deepmind-research-engineer-interview> · ML-From-Scratch <https://github.com/eriklindernoren/ML-From-Scratch> · nn-zero-to-hero <https://github.com/karpathy/nn-zero-to-hero> · minGPT <https://github.com/karpathy/minGPT> · tinygrad <https://github.com/tinygrad/tinygrad> · LLMs-from-scratch (Raschka) <https://github.com/rasbt/LLMs-from-scratch>

Thread R — Research output, networking & applications · ~2 hrs/week (~110 hrs)

- **Reproduce-a-target-paper (monthly):** publicly reproduce one paper from a target team + write it up. <https://paperswithcode.com/> · How to Read a Paper (Keshav) <https://web.stanford.edu/class/ee384m/Handouts/HowtoReadPaper.pdf> · Connected Papers <https://www.connectedpapers.com/> · Papers We Love <https://github.com/papers-we-love/papers-we-love>
- **Write:** 1-2 short posts/month; "why me for digital biology" narrative + two job-talk decks.
- **Publish:** aim the spike capstone at a venue — MICCAI <https://www.miccai.org/> · MIDL <https://midl.io/> · MLSB <https://www.mlsb.io/> · LoG <https://logconference.org/> · ISMB/RECOMB <https://www.iscb.org/> · arXiv <https://arxiv.org/> · bioRxiv <https://www.biorxiv.org/> · OpenReview <https://openreview.net/>
- **Applications:** run the company ladder; from Tier 0 this thread carries your live loops.

BLOCK 1 — Foundations, domain, imaging-video & the spike (*done ~Apr 2027*)

PHASE 0 — Fundamentals Rebuild: Stats, ML & DL · ~55 hrs · Aug 2026 · do first → then start applying

Rebuild crisp, *out-loud* command + vocabulary; intuition-first; **build your living glossary as you go**. Primary — StatQuest <https://www.youtube.com/@statquest/playlists> · <https://statquest.org/>

- **0.1 Statistics (~14 hrs)** — distributions; variance/SD; CLT; sampling; hypothesis testing & p-values; CIs; R^2 ; covariance/correlation; MLE; bootstrapping; power.
- **0.2 Classic ML (~20 hrs)** — bias-variance; CV; confusion matrix, sensitivity/specificity, precision/recall, ROC/AUC (& when PR is honest); linear/logistic regression; ridge vs lasso & why L1→sparsity; trees/RF/AdaBoost/GBM/XGBoost; SVMs & kernels; k-means/hierarchical; PCA; naive Bayes; entropy & MI.
- **0.3 DL fundamentals (~14 hrs)** — StatQuest NN + 3B1B NN https://www.youtube.com/playlist?list=PLZHQBOWTQDNU6R1_67000Dx_ZCJB-3pi
- **0.4 Deliverable (~7 hrs)** — articulation cheat-sheet + glossary v1. Reference: StatQuest *Illustrated Guide to ML*.

Self-test (Feynman gate): explain 15 concepts out loud, plain language, no notes.

PHASE 1 — Mathematics + Neural Networks from First Principles · ~210 hrs · Sep - late Oct 2026 · a genuine build, do NOT rush

Real mathematical maturity — the thing that breaks the vanilla ceiling. **Visual → rigorous → derive by hand**, spaced.

- **1.1 Linear algebra (~72 hrs)** — 3B1B LA https://www.youtube.com/playlist?list=PLZHQBOWTQDPD3MizzM2xVFitgF8hE_ab · MIT 18.06 https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/video_galleries/video-lectures/ · MIT 18.065 <https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/> · Matrix Cookbook <https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf> · Strang *Intro to Linear Algebra* · Imperial Math-for-ML LA <https://www.coursera.org/learn/linear-algebra-machine-learning>
- **1.2 Calculus + matrix calculus (~26 hrs)** — 3B1B Calc <https://www.youtube.com/playlist?list=PLZHQBOWTQDMsr9K-rj53DwVRMYO3t5Yr> · Parr & Howard <https://explained.ai/matrix-calculus/> · Imperial Multivariate Calc <https://www.coursera.org/learn/multivariate-calculus-machine-learning>
- **1.3 Probability & statistics (~50 hrs)** — Stat 110 <https://projects.iq.harvard.edu/stat110/youtube> · MIT RES.6-012 <https://ocw.mit.edu/cours>

es/res-6-012-introduction-to-probability-spring-2018/ · Blitzstein & Hwang (free) <http://probabilitybook.net/>

- **1.4 Neural networks from first principles (~62 hrs) · PROTECT** — Karpathy Zero to Hero (all 10, re-implement each) <https://www.youtube.com/playlist?list=PLAqhlrjkbWl23v9cThsA9GvCAUhRvKZ> · hub <https://karpathy.ai/zero-to-hero.html> · repo <https://github.com/karpathy/nn-zero-to-hero>

Self-test (gate → Threads start): hand-derive backprop (2-layer MLP); micrograd from memory; why BatchNorm helps & what breaks without it; whiteboard attention + derive ∇d_k .

PHASE 2 — Applied CV & Medical Image Segmentation · ~195 hrs (+ Capstone 1 ~25) · late Oct - early Dec 2026 · → Tier 0/1

Turn what you *do* into what you can *explain and improve*. Apply to Tier-0 India roles in parallel.

- **2.1 Imaging physics (~30 hrs)** — Elster <https://mriquestions.com/index.html> · OHBM <https://www.youtube.com/@ohbmonline/playlists> · Callaghan https://www.youtube.com/playlist?list=PLqEMVgX2VgZcG_xR1QRdsuc-X_zEgMx-Z · iBiology <https://www.youtube.com/playlist?list=PLF513KEDjY9YBNhwMfX6jMI3PlpdW9TMP>
- **2.2 CNNs + conv math + U-Net + losses (~50 hrs)** — CS231n <https://www.youtube.com/playlist?list=PL3FW7Lu3i5vHM8ljYj-zLqRF3EO8sYv> · <https://cs231n.github.io/> · conv arithmetic <https://arxiv.org/abs/1603.07285> · U-Net <https://arxiv.org/abs/1505.04597> · V-Net <https://arxiv.org/abs/1606.04797> · Attention U-Net <https://arxiv.org/abs/1804.03999> · nnU-Net <https://www.nature.com/articles/s41592-020-01008-z> · loss survey <https://arxiv.org/abs/2006.14822> · Generalized Dice <https://arxiv.org/abs/1707.03237> · Focal Tversky <https://arxiv.org/abs/1810.07842> · Boundary <https://arxiv.org/abs/1812.07032> (implement from scratch — Thread M)
- **2.3 nnU-Net + MONAI + microscopy (~45 hrs)** — MONAI <https://github.com/Project-MONAI/tutorials> · <https://www.youtube.com/@projectmonai/videos> · nnU-Net v2 <https://github.com/MIC-DKFZ/nnUNet> · Kitware <https://www.kitware.com/developing-custom-3d-medical-image-segmentation-solutions-using-out-of-the-box-pipelines-in-monai/> · DigitalSreeni <https://www.youtube.com/@DigitalSreeni/playlists> · code https://github.com/bnsreenu/python_for_microscopists · AI for Medical Diagnosis <https://www.coursera.org/learn/ai-for-medical-diagnosis>
- **2.4 Transformer segmentation + foundation models (~20 hrs)** — ViT <https://arxiv.org/abs/2010.11929> · TransUNet <https://arxiv.org/abs/2102.04306> · Swin-UNet <https://arxiv.org/abs/2105.05537> · UNETR <https://arxiv.org/abs/2103.10504> · Swin UNETR <https://arxiv.org/abs/2201.01266> · SAM <https://arxiv.org/abs/2304.02643> · MedSAM <https://www.nature.com/articles/s41467-024-44824-z> · MedNeXt <https://arxiv.org/abs/2303.09975>
- **2.5 3D, uncertainty, evaluation (~40 hrs)** — TorchIO <https://torchio.readthedocs.io/> · Yarin Gal uncertainty https://www.youtube.com/result?search_query=yarin+gal+uncertainty+deep+learning · Metrics Reloaded <https://www.nature.com/articles/s41592-023-02151-z> · Stanford AIMI <https://aimi.stanford.edu/education/educational-resources>

CAPSTONE 1 (~25 hrs) · public data · Tier-0/1 gate. Preprocessing (physics-justified), architecture ablation (3), loss ablation, uncertainty + failure modes; implement losses + one block from scratch. Datasets: Medical Decathlon <http://medicaldecathlon.com/> · BraTS <https://www.synapse.org/brats> · TCIA <https://www.cancerimagingarchive.net/> · IXI <https://brain-development.org/ixi-dataset/> · LIVECell <https://github.com/sartorius-research/LIVECell> · Sartorius <https://www.kaggle.com/competitions/sartorius-cell-instance-segmentation> · BBBC <https://bbbc.broadinstitute.org/> **Self-test:** whiteboard U-Net + skips with shapes; why nnU-Net beats fancier architectures (and when not); diagnose bias-field/imbalance/label-noise/domain-shift.

PHASE 2V — Imaging & Video extension (life-sciences-native) · ~70 hrs (+ Capstone 2) · Dec 2026 · → Tier 1 (pulled into Block 1 for the ladder)

Extend your imaging leg into video/temporal — unlocks phenomics/cell-imaging (Recursion) and strengthens the India case.

- **2V.1 Video & temporal (~40 hrs)** — VideoMAE <https://arxiv.org/abs/2203.12602> · TimeSformer <https://arxiv.org/abs/2102.05095> · ViViT <http://arxiv.org/abs/2103.15691> · SlowFast <https://arxiv.org/abs/1812.03982> · SAM 2 <https://arxiv.org/abs/2408.00714> · <https://github.com/facebookresearch/sam2> · RAFT <https://arxiv.org/abs/2003.12039> · ByteTrack <https://arxiv.org/abs/2110.06864> · OC-SORT <https://arxiv.org/abs/2203.14360> · BoT-SORT <https://arxiv.org/abs/2206.14651> · HOTA <https://arxiv.org/abs/2009.07736> · MOTChallenge <https://motchallenge.net/> · DAVIS <http://davischallenge.org/> · YouTube-VOS <https://youtube-vos.org/>
- **2V.2 Life-sciences-native video (~30 hrs) — the differentiator** — DeepLabCut <https://www.deeplabcut.org/> · <https://github.com/DeepLabCut/DeepLabCut> · <https://www.nature.com/articles/s41593-018-0209-y> · SLEAP <https://sleap.ai/> · <https://www.nature.com/articles/s41592-022-01426-1> · Cellpose <https://www.cellpose.org/> · TrackMate <https://imagej.net/plugins/trackmate/> · Ultracx <https://github.com/royerlab/ultrack> · Cell Tracking Challenge <http://celltrackingchallenge.net/>

CAPSTONE 2 (video) · public data · Tier 1. DeepLabCut/SLEAP pose → temporal model → behavior classification; or Cellpose seg → TrackMate/Ultracx tracking + lineage; SAM 2 for mask propagation. Datasets: CalMS21 <https://data.caltech.edu/records/s0vdx-0k302> · MABe <https://www.aicrowd.com/challenges/multi-agent-behavior-challenge-2022> · Cell Tracking Challenge <http://celltrackingchallenge.net/> **Self-test:** SAM 2's memory mechanism; tracking-by-detection vs joint; when a video transformer beats frame-wise + tracking; cell-lineage failure modes.

PHASE 3 — DL Theory & Modern Architectures · ~165 hrs · Jan 2027 · → Tier 1 (broad)

Real modeling intuition — optimization dynamics, information theory, modern architectures, generative/diffusion.

- **3.1 Optimization (~30 hrs)** — Ruder <https://www.ruder.io/optimizing-gradient-descent/> · Boyd EE364A <https://www.youtube.com/playlist?list=PL3940DD956CDF0622> · <https://web.stanford.edu/~boyd/cvxbook/>
- **3.2 Information theory (~20 hrs)** — MacKay ITILA <https://www.inference.org.uk/itila/book.html>
- **3.3 Transformers in depth + variants (~44 hrs)** — CS25 <https://web.stanford.edu/class/cs25/> · https://www.youtube.com/playlist?list=PLoROMvovd4rNijRchCzutFw5ltR_Z27CM · Lilian Weng <https://lilianweng.github.io/> · Annotated Transformer <http://nlp.seas.harvard.edu/annotated-transformer/> · RoPE <https://arxiv.org/abs/2104.09864> · RMSNorm <https://arxiv.org/abs/1910.07467> · FlashAttention <https://arxiv.org/abs/2205.14135> · GQA <https://arxiv.org/abs/2305.13245> · SwiGLU <https://arxiv.org/abs/2002.05202> · MoE <https://arxiv.org/abs/2101.03961>
- **3.4 ViT + modern ConvNets (~27 hrs)** — Lucas Beyer https://www.youtube.com/results?search_query=lucas+beyer+vision+transformer · ConvNeXt <https://arxiv.org/abs/2201.03545> · DINOv2 <https://arxiv.org/abs/2304.07193> · MAE <https://arxiv.org/abs/2111.06377> · HF CV <https://huggingface.co/learn/computer-vision-course>
- **3.5 Generative & diffusion (~44 hrs) · PROTECT — feeds the spike** — CS236 <https://www.youtube.com/playlist?list=PLoROMvovd4rPOW>

A-omMM6STXaWW4FvJT8 · DDPM <https://arxiv.org/abs/2006.11239> · Yang Song <https://yang-song.net/blog/2021/score/> · Lilian Weng diffusion ht <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/> · HF Diffusion <https://huggingface.co/learn/diffusion-course>

Self-test: derive the diffusion loss (variational + score-matching); compare positional encodings; SSL in low-data; what momentum/warmup/schedules change in training dynamics.

PHASE 5 — Production AI Systems · ~165 hrs · Feb 2027

- **5.1 LLM internals (~34)** — CS336 <https://stanford-cs336.github.io/spring2024/> · https://www.youtube.com/playlist?list=PLoROMvovd4rOY23Y0BoGoBGgQ1zmU_MT_ · HF NLP <https://huggingface.co/learn/nlp-course>
- **5.2 RAG (~28)** — Hamel <https://www.oreilly.com/radar/what-we-learned-from-a-year-of-building-with-llms-part-i/> · Eugene Yan <https://eugeneyan.com/writing/llm-patterns/> · Anthropic <https://www.anthropic.com/news/contextual-retrieval> · LlamaIndex <https://docs.llamaindex.ai/> · RAGAS <https://docs.ragas.io/> · DSPy <https://dspace.ai/>
- **5.3 MLOps + ML systems design (~44) · KEEP** — Chip Huyen <https://huyenchip.com/ml-interviews-book/contents/8.1.1-ml-systems-design.html> · CS329S <https://stanford-cs329s.github.io/> · Made With ML <https://madewithml.com/> · Full Stack DL <https://fullstackdeeplearning.com/course/2022/>
- **5.4 Distributed training + inference (~26)** — Raschka <https://magazine.sebastianraschka.com/> · HF multi-GPU https://huggingface.co/docs/transformers/en/perf_train_gpu_many · vLLM <https://arxiv.org/abs/2309.06180> · ZeRO <https://arxiv.org/abs/1910.02054>
- **5.5 Eval/observability/safety + systems-design prep (~33)** — Anthropic Constitutional AI <https://www.anthropic.com/news/constitutional-ai> · Eugene Yan evals <https://eugeneyan.com/writing/llm-evaluators/> · observability (LangSmith/Phoenix/Helicone); ML-systems-design whiteboards.

PHASE 6 — Geometric / Molecular ML — THE SPIKE · ~185 hrs (+ Capstone 3 ~35, flagship) · Mar - mid-Apr 2027 · → Tier 2 · BLOCK 1 COMPLETE

The differentiator for the relocation target + PhD seed. Your growing QML understanding makes the quantum-informed angle uniquely credible, on public benchmarks.

- **6.1 GDL + graph ML (~50)** — CS224W <https://web.stanford.edu/class/cs224w/> · 2023 <https://snap.stanford.edu/class/cs224w-2023/> · <https://www.youtube.com/playlist?list=PLoROMvovd4rPLKxlpqjhPgDQy7imNkDn> · Bronstein <https://geometricdeeplearning.com/lectures/> · proto-book <https://arxiv.org/abs/2104.13478> · Veličković https://www.youtube.com/results?search_query=petar+velickovic+gnn+lectures · PyG <https://pytorch-geometric.readthedocs.io/> · colabs https://pytorch-geometric.readthedocs.io/en/latest/get_started/colabs.html · DGL <https://www.dgl.ai/>
- **6.2 Molecular ML + equivariant nets (~55) · core** — DeepChem <https://deepchem.io/tutorials/> · TDC <https://tdcommons.ai/> · SchNet <https://arxiv.org/abs/1706.08566> · DimeNet++ <https://arxiv.org/abs/2011.14115> · GemNet <https://arxiv.org/abs/2106.08903> · MolFormer <https://arxiv.org/abs/2106.09553> · Cohen&Welling <https://arxiv.org/abs/1602.07576> · EGNN <https://arxiv.org/abs/2102.09844> · e3nn <https://e3nn.org/> · <https://github.com/e3nn/e3nn> · tutorial https://blondegeek.github.io/e3nn_tutorial/ · NequIP <https://github.com/mir-group/nequip> · <https://www.nature.com/articles/s41467-022-29939-5> · MACE <https://github.com/ACEsuit/mace> · <https://arxiv.org/abs/2206.07697> · Allegro <https://github.com/mir-group/allegro> · Equiformer <https://arxiv.org/abs/2206.11990> · curated list <https://github.com/Eipgen/Neural-Network-Models-for-Chemistry>
- **6.3 Protein structure + equivariance (~27)** — AlphaFold 2 (AlQuraishi) https://www.youtube.com/results?search_query=AlphaFold+2+explained+AlQuraishi · <https://www.nature.com/articles/s41586-021-03819-2> · ESM/ESMFold <https://github.com/facebookresearch/esm> · ESM Atlas <https://esmatlas.com/> · Boltz <https://github.com/jwohlwend/boltz> · OpenFold <https://github.com/aqlaboratory/openfold>
- **6.4 JAX + mock interviews + portfolio (~28)** — 45-min segmentation deep-dive; 30-min behavioral; **translate a capstone component to JAX** <https://jax.readthedocs.io/> · Equinox <https://docs.kidger.site/equinox/> · Flax <https://flax.readthedocs.io/>

CAPSTONE 3 (~35, flagship) · public quantum-chemistry data · unlocks Tier 2. E(3)-equivariant GNN (SchNet → DimeNet++ → EGNN/NequIP) predicting molecular properties; ablate equivariance; **fold in the quantum-informed angle** (delta-ML toward CCSD; DFT-vs-quantum data-generation → connects to Phase Q). Implement message-passing from scratch; translate a component to JAX. Datasets: QM9 <https://quantum-machine.org/datasets/> · PCQM4Mv2 <https://ogb.stanford.edu/kddcup2021/pcqm4m/> · MoleculeNet <https://moleculenet.org/> · OGB <https://ogb.stanford.edu/> · TDC <https://tdcommons.ai/>. **The flagship — do it well** → MLSB/AI4Science/LoG workshop paper (Thread R). **Self-test:** message passing vs convolution; what equivariance buys & why it matters for molecules; when a GNN beats a fingerprint baseline; AlphaFold's structure-module equivariance; the quantum-chemistry data-generation story end-to-end. → **BLOCK 1 COMPLETE** → **Tier 2 RE interview-ready.**

PHASE Q — Quantum Computing & QML (full-depth track) · ~305 hrs (+ Capstone Q ~30) · run parallel from ~Apr 2027, concentrated ~Aug-Nov 2027 · done ~end-2027

Your quantum exposure was overseeing a vendor (QpiAI's retrosynthesis work), not hands-on depth, and you want real QML expertise — so this is a complete build at full v6 hours + added resources. **Run it parallel-lite (a few hrs/wk) alongside Block 2 from ~Apr, then as a concentrated block ~Aug-Nov once Block 2's phases are done** — off the job-critical path, so full depth never delays your ladder. It seeds a QML career branch (QpiAI) and deepens the spike.

Intuition-first entry (before the rigorous courses): Quantum Country (spaced-repetition primer) <https://quantum.country/qcvc> · MIT 8.04 (Adams) https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/video_galleries/lecture-videos/

- **Q.1 Group theory & symmetry (~22)** — group theory for physicists https://www.youtube.com/results?search_query=group+theory+for+physicists+lectures · GDL <https://geometricdeeplearning.com/lectures/>
- **Q.2 QM math (~25)** — MIT 8.04 <https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/> · Quantum Country <https://quantum.country/qcvc> · (opt) Schuller https://www.youtube.com/playlist?list=PLPH7f_7ZlzxTi6kS4vCmv4ZKm9u8g5yic
- **Q.3 QC fundamentals — Watrous/IBM (~80) · rigorous core** — Basics <https://learning.quantum.ibm.com/course/basics-of-quantum-information> · <https://www.youtube.com/playlist?list=PLOFEBzvs-VvrXTMy5Y2lqmSaUjfnhvBHR> · Algorithms <https://learning.quantum.ibm.com/course/fundamentals-of-quantum-algorithms> · General Formulation <https://learning.quantum.ibm.com/course/general-formulation-of-quantum-information> · Error Correction <https://learning.quantum.ibm.com/course/foundations-of-quantum-error-correction> · Nielsen & Chuang ch 1-4,8,10 ·

- Qiskit/IBM hub <https://learning.quantum.ibm.com/> · Preskill Ph219 <http://theory.caltech.edu/~preskill/ph219/> · PennyLane Codebook <https://pennylane.ai/codebook> · IBM Quantum Challenges <https://www.ibm.com/quantum>
- **Q.4 VQE, ADAPT-VQE, DMET, QAOA (~45)** — PennyLane chemistry https://pennylane.ai/qml/demos_quantum-chemistry · VQE <https://www.nature.com/articles/ncomms5213> · ADAPT-VQE <https://www.nature.com/articles/s41467-019-10988-2> · DMET-VQE <https://arxiv.org/abs/2108.08987> · QAOA <https://arxiv.org/abs/1411.4028> · Hadfield <https://arxiv.org/abs/1709.03489> · Pauli grouping <https://arxiv.org/abs/1907.13117>
 - **Q.5 QML theory (~42) — the honest-take material** — PennyLane QML (Schuld) <https://pennylane.ai/topics/quantum-machine-learning> · Schuld "Taking stock" https://www.youtube.com/results?search_query=maria+schuld+taking+stock+quantum+machine+learning · Schuld & Petruccione ch 4-7 · QML=kernels <https://arxiv.org/abs/2101.11020> · Barren plateaus <https://www.nature.com/articles/s41467-018-07090-4> · Expressibility <https://arxiv.org/abs/1905.10876> · Advantage-goal <https://arxiv.org/abs/2203.01340> · PQC encodings <https://arxiv.org/abs/2008.08605> · Péré <https://github.com/Christophe-pere/QML-Course> · Hjorth-Jensen <https://github.com/CompPhysics/QuantumComputingMachineLearning> · TensorFlow Quantum <https://www.tensorflow.org/quantum> · PennyLane demos <https://pennylane.ai/qml/demonstrations/>
 - **Q.6 Quantum chemistry on quantum computers + retrosynthesis context (~60)** — Qiskit GSS <https://www.youtube.com/@qiskit/playlists> · IBM Quantum Learning + SQD/Krylov <https://learning.quantum.ibm.com/> · <https://www.ibm.com/quantum/blog/iql-migration>

CAPSTONE Q (~30) · public molecules. H₂/LiH/H₂O or a QM9 subset (**not** Lilly molecules; clear of QpAI-engagement specifics). Vanilla VQE + UCCSD, ADAPT-VQE, DMET-VQE, benchmarked vs classical CCSD; write-up defending ansatz/optimizer/measurement grouping/noise mitigation/basis set + a barren-plateau analysis. Optional delta-ML surrogate → connects to Capstone 3. **Self-test:** derive VQE (variational principle); parameter-shift rule & why it works; barren-plateau argument; VQE vs QAOA — when each & when *no* quantum algorithm; honest QML hype-vs-promise (cite Schuld); DMET fragmentation & the classical-quantum interface. → **QML career branch opens.**

BLOCK 2 — CV-Expert breadth, GenAI & Interview Crescendo (*done ~Jul-Aug 2027*)

Threads T + M + R continue; Phase Q runs parallel-lite.

PHASE 7 — Generative AI & Agentic AI · ~155 hrs (+ Capstone 4) · Apr - May 2027

- **7.1 Advanced LLMs (~42)** — LoRA <https://arxiv.org/abs/2106.09685> · QLoRA <https://arxiv.org/abs/2305.14314> · Raschka <https://magazine.sebastianraschka.com/> · RLHF <https://arxiv.org/abs/2203.02155> · DPO <https://arxiv.org/abs/2305.18290> · HF LLM course <https://huggingface.co/learn/nllm-course> · Chip Huyen *AI Engineering*
- **7.2 Multimodal/VLM (~26)** — CLIP <https://arxiv.org/abs/2103.00020> · LLaVA <https://arxiv.org/abs/2304.08485> · Flamingo <https://arxiv.org/abs/2204.14198>
- **7.3 Agents (~52)** — HF Agents <https://huggingface.co/learn/agents-course> · HF MCP <https://huggingface.co/learn/mcp-course> · Anthropic <https://www.anthropic.com/research/building-effective-agents> · Andrew Ng <https://www.deeplearning.ai/short-courses/> · LangGraph <https://langchain-ai.github.io/langgraph/> · **CAPSTONE 4 · public literature.** Multi-step research agent (RAG + tools + memory + eval) over **open** literature. PMC-OA <https://www.ncbi.nlm.nih.gov/pmc/tools/openftlist/> · arXiv <https://arxiv.org/> · PubMed E-utilities <https://www.ncbi.nlm.nih.gov/books/NBK25501/> · Semantic Scholar API <https://www.semanticscholar.org/product/api>. **Never ingest Lilly internal data.**

PHASE 8 — Computer Vision Expert (classical / geometric / 3D / edge) · ~175 hrs (+ Capstone 5) · May - Jun 2027

Round out into a broad CV expert so no vision question catches you flat (video already done in Phase 2V).

- **8.1 Classical & geometric CV (~45)** — FPCV (Nayar) <https://fpcv.cs.columbia.edu/> · Coursera <https://www.coursera.org/specializations/firstprinciplesofcomputervision> · Stachniss <https://www.youtube.com/@CyrillStachniss> · Hartley & Zisserman · Szeliski (free) <https://szeliski.org/Book/> · COLMAP <https://colmap.github.io/> · OpenMVG <https://github.com/openMVG/openMVG> · ORB-SLAM3 https://github.com/UZ-SLAMLab/ORB_SLAM3 · Open3D <http://www.open3d.org/>
- **8.2 Modern DL vision breadth (~35)** — Michigan EECS 498 <https://web.eecs.umich.edu/~justincj/teaching/eecs498/> · <https://www.youtube.com/playlist?list=PL5-TkQAFzbzxBHtZdVCWE0Zbhong7r> · Detectron2 <https://github.com/facebookresearch/detectron2> · MMDetection <https://github.com/open-mmlab/mmdetection> · Ultralytics YOLO <https://docs.ultralytics.com/>
- **8.5 Deployment & optimization for hardware/edge (~50)** — OpenCV <https://docs.opencv.org/> · PyImageSearch <https://pyimagesearch.com/> · ONNX <https://onnx.ai/> · TensorRT <https://developer.nvidia.com/tensorrt> · PMPP (Kirk & Hwu) · NVIDIA DLI <https://www.nvidia.com/en-us/training/> · Triton <https://github.com/triton-inference-server/server> · DeepStream <https://developer.nvidia.com/deepstream-sdk> · CUDA Guide <https://docs.nvidia.com/cuda/cuda-c-programming-guide/>
- **8.6 3D vision & neural rendering (~30)** — NeRF <https://arxiv.org/abs/2003.08934> · 3D Gaussian Splatting <https://arxiv.org/abs/2308.04079> · nerfstudio <https://docs.nerf.studio/> · Open3D point clouds <http://www.open3d.org/docs/release/tutorial/geometry/pointcloud.html> · **CAPSTONE 5 (deployment) · public data.** SfM/calibration/visual-odometry on public sets, **or** take C1/C2's model → optimized C++ inference (TensorRT/ONNX, quantized). Datasets: KITTI <https://www.cvlibs.net/datasets/kitti/> · TUM RGB-D <https://cvg.cit.tum.de/data/datasets/rgbd-datas> · ETH3D <https://www.eth3d.net/> · **Self-test:** pinhole model + epipolar constraint; SIFT & scale/rotation invariance; two-stage vs one-stage detectors; what TensorRT does; porting a PyTorch CV model to a real-time C++ pipeline.

PHASE 9 — Research + Big-Tech Interview Crescendo · ~160 hrs · Jun - ~Aug 2027 · → Tier 3 · BLOCK 2 COMPLETE

RE loop = 2 medium-hard coding + ML fundamentals + ML system design + behavioral (+ research discussion for research roles).

- **9.1 DSA — RE weight (~55)** — timed NeetCode 150 → Blind 75 → company-tagged <https://neetcode.io/> · <https://leetcode.com/> · DesignGurus <https://www.designgurus.io/> · Striver <https://takeuforward.org/>. Blank editor, ~35 min, code must run; ~4-6/day.
- **9.2 Implement-by-hand ML (~35) · highest value** — timed, unaided, blank file: attention, a loss, a sampler, a diffusion step, a training loop, backprop, a GNN layer; + ML math/theory sweep (gradients, KL, sampling, convergence, "L1 sparsity via gradients").
- **9.3 System design (~35)** — ML system design first; ByteByteGo <https://bytebytego.com/> · <https://www.youtube.com/@ByteByteGo> · Gaurav Sen <https://www.youtube.com/@gkcs> · DDIA + MIT 6.824 <https://pdos.csail.mit.edu/6.824/> for loops with generic system design. 8-10 prompts,

recorded, self-critiqued.

- **9.4 Research job-talk + behavioral (~35)** — 20-min talk + Q&A on your two strongest capstones; ~6–8 STAR stories per org; low-level/OO design where the loop includes it; live mocks (Pramp / interviewing.io). **Research loops ban AI tools — rehearse unaided. Self-test:** a random LeetCode hard in ~45 min, clean running code; implement attention + a loss + a sampler + a GNN layer unaided, timed; a full system-design whiteboard in 45 min; a 20-min job-talk; four STAR stories from memory. → **BLOCK 2 COMPLETE** → **Tier 3 top-lab loops**.

(Active interviewing (~Aug → Dec 2027)): keep all threads warm ~5 hrs/wk; run Phase Q concentrated in parallel; buffer for on-sites, take-homes, rest.)

Application overlay & company target ladder

Apply from ~month 2 on your existing applied skill; climb as each capstone lands. "Relocate" = outside India.

Tier	Unlocked by	~When	Apply to
Tier 0	Applied skill + Capstone 1	~Dec 2026 (apply from ~Sep)	India medical-imaging AI + accessible; stretch Google-India
Tier 1	+ Capstone 2 (video)	~Jan 2027	broader medical/biomedical imaging + video; Recursion/phenomics; Google-India
Tier 2	+ Capstone 3 (spike)	~Apr 2027	AI drug-discovery labs; Isomorphic / DeepMind-science RE
Tier 3	+ interview crescendo	~Jul-Aug 2027	full top-lab loops
QML branch	+ Capstone Q	~Nov 2027	quantum-for-science roles

Tier 0: Qure.ai <https://qure.ai/> · SigTuple <https://sigtuple.com/> · NVIDIA <https://www.nvidia.com/en-us/about-nvidia/careers/> · GE HealthCare <https://www.gehealthcare.com/> · MSR India <https://www.microsoft.com/en-us/research/lab/microsoft-research-india/> · Google Research India <https://research.google/locations/india/> **Tier 1:** Aidoc, Rad AI, Annalise.ai, PathAI, Paige.AI, Owkin <https://www.owkin.com/> **Tier 2 (relocate):** Recursion (phenomics = DL on cell images — direct match) <https://www.recursion.com/careers> · Insitro <https://insitro.com/> · Genesis <https://www.genesisltherapeutics.ai/> · Iambic <https://www.iambic.ai/> · Chai <https://www.chaidiscovery.com/> · EvolutionaryScale <https://www.evolutionaryscale.ai/> · Profluent <https://www.profluent.bio/> · InstaDeep <https://www.instadeep.com/> · Valence Labs <https://www.valencelabs.com/> · **Isomorphic Labs** <https://www.isomorphiclabs.com/careers> · **Google DeepMind** <https://deepmind.google/about/careers/>. Target Research Engineer / ML Research Engineer (MSc + ~2 yrs, no PhD). Isomorphic has a Lilly partnership — not a blocker, apply with normal discretion. **QML branch:** QpiAI (Bengaluru; QpiAI-Pharma; quantum-chem / quantum-ML / AI-native roles) <https://www.qpi.ai/tech/careers> · IBM Quantum <https://www.ibm.com/quantum> **Tier 3:** Isomorphic, Google DeepMind, Google Research, Genentech gRED / Prescient Design, Microsoft Research Health Futures, Meta FAIR.

Conflict-of-interest guardrails

Not legal advice — verify with the right people at Lilly.

1. **Public data only.** Never Lilly proprietary data/targets/molecules/assays/docs. In Capstone Q/3 use public molecules (H₂/LiH/H₂O/QM9), **not** Lilly molecules (avoid keto-enol / HATU work), and **clear of QpiAI-engagement specifics**.
2. **Generic, published methods** only — no Lilly/vendor trade secrets.
3. **Personal time & resources** — your own hardware/accounts. (Lilly-hardware/network work must use Lilly's JFrog Artifactory per policy; these capstones are personal-hardware/personal-account.)
4. **Clear before you publish** — Lilly's external-publication / IP review; when unsure, ask first.
5. **Check your agreement** — employment / invention-assignment / moonlighting terms.
6. **Positioning stays generic** — transferable skills, never Lilly/vendor specifics.

Cross-cutting habits (weekly)

De-duplicated: old habits #3/#6/#7 live inside Threads M and R.

1. **Paper-of-the-week** — one paper, derivations redone on paper.
2. **Whiteboard Friday** — one concept, no notes; photograph → prep deck.
3. **Sunday writeup** — teach the week's deepest concept to a hypothetical junior.
4. **War chest + living glossary** — derivations, paper one-liners, gotchas, and every confusing term in your own words (Confusion Buffer #2).

PhD-later decision guide

Whether. RS at the target labs effectively needs a PhD + publications; RE accepts MSc + ~2 yrs (no PhD). RS title → PhD; senior/staff RE → optional. Your Phase-Q depth also opens a **quantum-ML / computational-chemistry PhD**. **When.** After landing a strong RE role — job-first expands options. **Where:** India part-time/external (IISc ERP <https://www.iisc.ac.in/admissions/external-registration-programme-ph-d/> — ~2 yrs' experience, ~1 yr on-campus, ~6 yrs; IITs external) · EU salaried/industrial (~€3–6k/mo, ~4 yrs; MSCA industrial doctoral networks; EURAXESS <https://euraxess.ec.europa.eu/>) — the strongest prestige-and-funding leap · USA mostly full-time funded (research-embedded, or skip via RE→staff). **Set-up now (zero marginal cost):** spike capstone → workshop paper + preprint; 1–2 advisor relationships; Capstone Q gives a second (QML) research identity. **Decision criteria:** advisor fit → topic continuity → funding model → part-time feasibility → location/visa → prestige (last).

Publication & portfolio strategy

Realistic over ~18-24 mo: **1-2 non-archival workshop papers + 1 preprint + clean open-source repos.**

- **Job lever:** reproduce-a-paper + a workshop paper = paper-discussion prep, referral hook, proof of level. MLSB prioritizes **open-sourced code** for spotlights.
- **PhD lever:** first-author workshop papers + preprints + an advisor relationship lift you to funded EU/industrial or research-embedded routes.
- **Venues:** imaging/video → MIDL/MICCAI workshop; molecular → MLSB/LoG/AI4Science; QML → a quantum-ML workshop or arXiv. Target **MIDL-2027 / MLSB-2027 + a rolling bioRxiv/arXiv preprint.**
- **Visibility:** reproducible READMEs; a merged PR into an ecosystem repo (**e3nn / MACE / PyG / MONAI / PennyLane**) beats a blog post.

Risk register

Risk	Like liho od	Im pa ct	Leading indicator	Mitigation
Confusion compounding — <i>your key risk</i>	High	Hi gh	Self-tests failed/skipped; a term keeps reappearing	Confusion Buffer — understanding-gated checkpoints, spaced re-derivation, consolidation weeks, triangulation. Don't advance until the Feynman gate passes
Overload / burnout (~2,530 hrs + full-time)	High	Hi gh	Skipped rest/consolidation weeks; capstone quality dropping; dread	Aggressive working weeks buffered by consolidation weeks; stretch the timeline, don't compress ; protect one rest day
Breadth-without-depth (full CV + full QML)	Med	Hi gh	Can use but can't derive/teach	Self-tests gate; Thread M + Whiteboard Friday enforce depth; the spike stays the priority for relocation
QML pulling focus off the job path	Med	Me d	Doing Phase Q before Tier-2	Locked to parallel/after-Tier-2
Over-claiming CV seniority	Med	Hi gh	Positioning says "senior CV engineer"; freeze on SLAM	You now have the depth — but keep "medical-imaging + geometric/molecular ML" primary, CV breadth as strong supporting
Goal drift / re-bloating	Med	Me d	Adding "just one more" course	Every addition serves imaging-video / molecular-geometric / the RE loop / (deliberately) QML — else cut
Relocation-vs-domestic split	Med	Me d	Effort split 50/50	Two legs share an umbrella; apply Tier-0 India <i>now</i> while building the Tier-2 spike
Premature / ill-fitting PhD	Low-Med	Hi gh	Applying to a self-funded tier-2/3 PhD before the job	PhD is post-job; job-first expands options

Resource index (quick lookup) — *all v6-v9 URLs preserved*

Brush-up (P0) — StatQuest <https://www.youtube.com/@statquest/playlists> · <https://statquest.org/> · 3B1B NN https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi

Foundations math (P1) — 3B1B LA https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab · 3B1B Calc <https://www.youtube.com/playlist?list=PLZHQObOWTQDMsr9K-rj53DwVRMYO3t5Yr> · MIT 18.06 <https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/> · MIT 18.065 <https://ocw.mit.edu/courses/18-065-matrix-methods-in-data-analysis-signal-processing-and-machine-learning-spring-2018/> · Stat 110 <https://projects.iq.harvard.edu/stat110/youtube> · MIT RES.6-012 <https://ocw.mit.edu/courses/res-6-012-introduction-to-probability-spring-2018/> · Matrix Cookbook <https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf> · Matrix Calculus <https://explained.ai/matrix-calculus/> · Karpathy Z2H <https://www.youtube.com/playlist?list=PLAqhlrjkxbuWI23v9cThsA9GvCAUhRvKZ> · hub <https://karpathy.ai/zero-to-hero.html> · Imperial LA <https://www.coursera.org/learn/linear-algebra-machine-learning> · Imperial Multivariate Calc <https://www.coursera.org/learn/multivariate-calculus-machine-learning> · Blitzstein & Hwang <http://probabilitybook.net/> · Z2H repo <https://github.com/karpathy/nn-zero-to-hero>

Later-threaded math — Boyd <https://www.youtube.com/playlist?list=PL3940DD956CDF0622> · <https://web.stanford.edu/~boyd/cvxbook/> · Ruder <https://www.ruder.io/optimizing-gradient-descent/> · MacKay <https://www.inference.org.uk/itila/book.html> · Group theory https://www.youtube.com/results?search_query=group+theory+for+physicists+lectures · GDL <https://geometricdeeplearning.com/lectures/>

Medical imaging (P2) — MRI Q&A <https://mrquestions.com/index.html> · OHBM <https://www.youtube.com/@ohbmonline/playlists> · Callaghan https://www.youtube.com/playlist?list=PLqEMVgX2VgZcG_xR1QRdsuc-X_Z_EgMx-Z · iBiology <https://www.youtube.com/playlist?list=PLF513KEDjY9YBNhwMfX6jMI3PdpW9TMP> · AI for Medical Diagnosis <https://www.coursera.org/learn/ai-for-medical-diagnosis> · DigitalSreeni <https://www.youtube.com/@DigitalSreeni/playlists> · code https://github.com/bnsreenu/python_for_microscopists · Conv arithmetic <https://arxiv.org/abs/1603.07285> · https://github.com/vdumoulin/conv_arithmetic · MONAI <https://github.com/Project-MONAI/tutorials> · <https://www.youtube.com/@projectmonai/videos> · nnUNet <https://github.com/MIC-DKFZ/nnUNet> · TorchIO <https://torchio.readthedocs.io/> · Stanford AIMI <https://aimi.stanford.edu/education/educational-resources> · CS231n <https://www.youtube.com/playlist?list=PL3FW7Lu3i5jvHM8ljYj-zLfQRF3EO8sYv> · <https://cs231n.github.io/> · Metrics Reloaded <https://www.nature.com/articles/s41592-023-02151-z> · U-Net <https://arxiv.org/abs/1505.04597> · V-Net <https://arxiv.org/abs/1606.04797> · Attention U-Net <https://arxiv.org/abs/1804.03999> · nnU-Net <https://www.nature.com/articles/s41592-020-01008-z> · Loss survey <https://arxiv.org/abs/2006.14822> · Generalized Dice <https://arxiv.org/abs/1707.03237> · Focal Tversky <https://arxiv.org/abs/1810.07842> · Boundary <https://arxiv.org/abs/1812.07032> · ViT <https://arxiv.org/abs/2010.11929> · TransUNet <https://arxiv.org/abs/2102.04306> · Swin-UNet <https://arxiv.org/abs/2105.05537> · UNETR <https://arxiv.org/abs/2103.10504> · Swin UNETR <https://arxiv.org/abs/2201.01266> · SAM <https://arxiv.org/abs/2304.02643> · MedSAM <https://www.nature.com/articles/s41467-024-44824-z> · MedNeXt <https://arxiv.org/abs/2303.09975> · **Datasets:** Medical Decathlon <http://medicaldecathlon.com/> · BraTS <https://www.synapse.org/brats> · TCIA <https://www.cancerimagingarchive.net/> · IXI <https://brain-development.org/ixi-dataset/> · LIVECell <https://github.com/sartorius-research/LIVECell> · Sartorius <https://www.kaggle.com/competitions/sartorius-cell-instance-segmentation> · BBBC <https://bbbc.broadinstitute.org/>

Deep learning core (P3) — CS25 <https://web.stanford.edu/class/cs25/> · https://www.youtube.com/playlist?list=PLoROMvody4rNijRchCzutFw5ItR_Z27CM · CS336 <https://stanford-cs336.github.io/spring2024/> · https://www.youtube.com/playlist?list=PLoROMvody4rOY23Y0BoGoBgGQ1zmU_MT_ · CS236 <https://www.youtube.com/playlist?list=PLoROMvody4rPOWA-omMM6STXaWW4FvJT8> · Lilian Weng <https://lilianweng.github.io/> · Annotated Transformer <http://nlp.seas.harvard.edu/annotated-transformer/> · HF NLP <https://huggingface.co/learn/nlp-course> · HF CV <https://huggingface.co/learn>

arn/computer-vision-course · HF Diffusion <https://huggingface.co/learn/diffusion-course> · RoPE <https://arxiv.org/abs/2104.09864> · RMSNorm <https://arxiv.org/abs/1910.07467> · FlashAttention <https://arxiv.org/abs/2205.14135> · GQA <https://arxiv.org/abs/2305.13245> · SwiGLU <https://arxiv.org/abs/2002.05202> · Switch/MoE <https://arxiv.org/abs/2101.03961> · ConvNeXt <https://arxiv.org/abs/2201.03545> · DINOv2 <https://arxiv.org/abs/2304.07193> · MAE <https://arxiv.org/abs/2111.06377> · DDPM <https://arxiv.org/abs/2006.11239> · Score-based <https://yang-song.net/blog/2021/score/> · Diffusion blog <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>

Quantum & QML (Phase Q) — IBM Quantum Learning <https://learning.quantum.ibm.com/> · Watrous <https://www.youtube.com/playlist?list=PLOFEBzvs-VvrXTMy5Y2IqmSaUjfnhvBHR> · Basics <https://learning.quantum.ibm.com/course/basics-of-quantum-information> · Algorithms <https://learning.quantum.ibm.com/course/fundamentals-of-quantum-algorithms> · General Formulation <https://learning.quantum.ibm.com/course/general-formulation-of-quantum-information> · Error Correction <https://learning.quantum.ibm.com/course/foundations-of-quantum-error-correction> · Qiskit <https://www.youtube.com/@qiskit/playlists> · QDA blog <https://www.ibm.com/quantum/blog/iql-migration> · PennyLane codebook <https://pennylane.ai/codebook> · PennyLane QML <https://pennylane.ai/topics/quantum-machine-learning> · PennyLane chemistry https://pennylane.ai/qml/demos_quantum-chemistry · MIT 8.04 <https://ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/> · Quantum Country <https://quantum.country/qvcv> · Schuller https://www.youtube.com/playlist?list=PLPH7f_7ZlzxTi6kS4vCmv4ZkM9u8g5yic · Péré <https://github.com/Christophe-pere/QML-Course> · Hjorth-Jensen <https://github.com/CompPhysics/QuantumComputingMachineLearning> · VQE <https://www.nature.com/articles/ncomms5213> · ADAPT-VQE <https://www.nature.com/articles/s41467-019-10988-2> · DMET-VQE <https://arxiv.org/abs/2108.08987> · QAOA <https://arxiv.org/abs/1411.4028> · Hadfield <https://arxiv.org/abs/1709.03489> · QML=kernels <https://arxiv.org/abs/2101.11020> · Barren plateaus <https://www.nature.com/articles/s41467-018-07090-4> · Expressibility <https://arxiv.org/abs/1905.10876> · Advantage-goal <https://arxiv.org/abs/2203.01340> · PQC encodings <https://arxiv.org/abs/2008.08605> · Pauli grouping <https://arxiv.org/abs/1907.13117> · Preskill Ph219 <http://theory.caltech.edu/~preskill/ph219/> · TensorFlow Quantum <https://www.tensorflow.org/quantum> · PennyLane demos <https://pennylane.ai/qml/demonstrations/> · IBM Quantum <https://www.ibm.com/quantum>

Production / MLOps (P5) — CS329S <https://stanford-cs329s.github.io/> · Made With ML <https://madewithml.com/> · Full Stack DL <https://fullstackdeeplearning.com/course/2022/> · Chip Huyen <https://huyenchip.com/ml-interviews-book/> · Hamel <https://www.oreilly.com/radar/what-we-learned-from-a-year-of-building-with-llms-part-i/> · Eugene Yan patterns <https://eugeneyan.com/writing/llm-patterns/> · evals <https://eugeneyan.com/writing/llm-evaluators/> · Anthropic retrieval <https://www.anthropic.com/news/contextual-retrieval> · Constitutional AI <https://www.anthropic.com/news/constitutional-ai> · LlamaIndex <https://docs.llamaindex.ai/> · RAGAS <https://docs.ragas.io/> · DSPy <https://dspy.ai/> · Raschka <https://magazine.sebastianraschka.com/> · HF multi-GPU https://huggingface.co/docs/transformers/en/perf_train_gpu_many · vLLM <https://arxiv.org/abs/2309.06180> · ZeRO <https://arxiv.org/abs/1910.02054>

DSA & C++ (Thread T) — learncpp <https://www.learncpp.com/> · The Chernobyl <https://www.youtube.com/@TheChernobyl> · cplusplus <https://en.cppreference.com/> · NeetCode <https://neetcode.io/> · LeetCode <https://leetcode.com/> · MIT 6.006 <https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/> · Abdul Bari https://www.youtube.com/@abdul_bari · DesignGurus <https://www.designgurus.io/> · Striver <https://takeuforward.org/> · Sean Prashad <https://seanprashad.com/leetcode-patterns/> · CP Handbook <https://cses.fi/book/book.pdf> · CSES <https://cses.fi/problemset/> · CppCon <https://www.youtube.com/@CppCon>

Implement-by-hand ML (Thread M) — d2l.ai <https://d2l.ai/> · labml.ai <https://nn.labml.ai/> · Annotated Transformer <http://nlp.seas.harvard.edu/annotated-transformer/> · micrograd <https://github.com/karpathy/micrograd> · nanoGPT <https://github.com/karpathy/nanoGPT> · DeepMind RE guide <https://igotanooffer.com/en/advice/google-deepmind-research-engineer-interview> · JAX <https://jax.readthedocs.io/> · Equinox <https://docs.kidger.site/equinox/> · Anki <https://apps.ankiweb.net/> · ML-From-Scratch <https://github.com/eriklindernoren/ML-From-Scratch> · nn-zero-to-hero <https://github.com/karpathy/nn-zero-to-hero> · minGPT <https://github.com/karpathy/minGPT> · tinygrad <https://github.com/tinygrad/tinygrad> · LLMs-from-scratch <https://github.com/rasbt/LLMs-from-scratch> · Flax <https://flax.readthedocs.io/>

Research output (Thread R) — Papers with Code <https://paperswithcode.com/> · MICCAI <https://www.miccai.org/> · MIDL <https://midl.io/> · MLSB <https://www.mlsb.io/> · LoG <https://logconference.org/> · ISCB/ISMB <https://www.iscb.org/> · arXiv <https://arxiv.org/> · bioRxiv <https://www.biorxiv.org/> · OpenReview <https://openreview.net/> · How to Read a Paper <https://web.stanford.edu/class/ee384m/Handouts/HowtoReadPaper.pdf> · Connected Papers <https://www.connectedpapers.com/> · Papers We Love <https://github.com/papers-we-love/papers-we-love>

GenAI & Agents (P7) — HF LLM course <https://huggingface.co/learn/llm-course> · HF Agents <https://huggingface.co/learn/agents-course> · HF MCP <https://huggingface.co/learn/mcp-course> · Anthropic Effective Agents <https://www.anthropic.com/research/building-effective-agents> · DeepLearning.AI <https://www.deeplearning.ai/short-courses/> · LangGraph <https://langchain-ai.github.io/langgraph/> · LoRA <https://arxiv.org/abs/2106.09685> · QLoRA <https://arxiv.org/abs/2305.14314> · RLHF <https://arxiv.org/abs/2203.02155> · DPO <https://arxiv.org/abs/2305.18290> · CLIP <https://arxiv.org/abs/2103.00020> · LLaVA <https://arxiv.org/abs/2304.08485> · Flamingo <https://arxiv.org/abs/2204.14198> · PMC-OA <https://www.ncbi.nlm.nih.gov/pmc/tools/openftl/> · PubMed E-utilities <https://www.ncbi.nlm.nih.gov/books/NBK25501/> · Semantic Scholar API <https://www.semanticscholar.org/product/api>

Computer Vision Expert + Video (P2V + P8) — FPCV <https://fpcv.cs.columbia.edu/> · Coursera <https://www.coursera.org/specializations/firstprinciplesofcomputervision> · Stachniss <https://www.youtube.com/@CyrilStachniss> · Michigan EECS 498 <https://web.eecs.umich.edu/~justincj/teaching/eeecs498/> · NeRF <https://arxiv.org/abs/2003.08934> · 3DGS <https://arxiv.org/abs/2308.04079> · OpenCV <https://docs.opencv.org/> · PyImageSearch <https://pyimagesearch.com/> · ONNX <https://onnx.ai/> · TensorRT <https://developer.nvidia.com/tensorrt> · VideoMAE <https://arxiv.org/abs/2203.12602> · TimeSformer <https://arxiv.org/abs/2102.05095> · ViViT <https://arxiv.org/abs/2103.15691> · SlowFast <https://arxiv.org/abs/1812.03982> · SAM 2 <https://arxiv.org/abs/2408.00714> · SAM <https://github.com/facebookresearch/sam2> · RAFT <https://arxiv.org/abs/2003.12039> · ViP <https://github.com/princeton-vl/RAFT> · ByteTrack <https://arxiv.org/abs/2110.06864> · iFZhang/ByteTrack · OC-SORT <https://arxiv.org/abs/2203.14360> · BoT-SORT <https://arxiv.org/abs/2206.14651> · HOTA <https://arxiv.org/abs/2009.07736> · MOTChallenge <https://motchallenge.net/> · DAVIS <https://davischallenge.org/> · YouTube-VOS <https://youtube-vos.org/> · DeepLabCut <https://www.deeplabcut.org/> · SLEAP <https://sleap.ai/> · Cellpose <https://www.cellpose.org/> · TrackMate <https://imagej.net/plugins/trackmate/> · Ultralytics <https://github.com/rogerlab/ultralytics> · Cell Tracking Challenge <http://celltrackingchallenge.net/> · CalMS21 <https://data.caltech.edu/records/s0vdx-0k302> · KITTI <https://www.cvlib.net/datasets/kitti/> · TUM RGB-D <https://cvg.cit.tum.de/data/datasets/rgbd-dataset> · ETH3D <https://www.eth3d.net/> · Szeliski <https://szeliski.org/Book/> · COLMAP <https://colmap.github.io/> · OpenMVG <https://github.com/openMVG/openMVG> · ORB-SLAM3 https://github.com/UR-ZSLAMLab/ORB_SLAM3 · Open3D <http://www.open3d.org/> · Detectron2 <https://github.com/facebookresearch/detectron2> · MMDetection <https://github.com/open-mmlab/mmdetection> · YOLO <https://docs.ultralytics.com/> · nerfstudio <https://docs.nerf.studio/> · Triton <https://github.com/triton-inference-server/server> · DeepStream <https://developer.nvidia.com/deepstream-sdk> · CUDA Guide <https://docs.nvidia.com/cuda/cuda-c-programming-guide/> · NVIDIA DLI <https://www.nvidia.com/en-us/training/>

Geometric / Molecular ML (P6 — the spike) — CS224W <https://web.stanford.edu/class/cs224w/> · Snap <https://snap.stanford.edu/class/cs224w-2023/> · PLoROM <https://www.youtube.com/playlist?list=PLoROMvodv4rPLKxlpqhjPgDQy7imNkDn> · Bronstein <https://geometricdeeplearning.com/lectures/> · proto-book <https://arxiv.org/abs/2104.13478> · PyG <https://pytorch-geometric.readthedocs.io/> · colabs https://pytorch-geometric.readthedocs.io/en/latest/get_started/colabs.html · DGL <https://www.dgl.ai/> · DeepChem <https://deepchem.io/tutorials/> · TDC <https://tdcommons.ai/> · SchNet <https://arxiv.org/abs/1706.08566> · DimeNet++ <https://arxiv.org/abs/2011.14115> · GemNet <https://arxiv.org/abs/2106.08903> · MolFormer <https://arxiv.org/abs/2106.09553> · Cohen&Welling <https://arxiv.org/abs/1602.07576> · EGNN <https://arxiv.org/abs/2102.09844> · e3nn <https://e3nn.org/> · e3nn <https://github.com/e3nn>

n/e3nn · tutorial https://blondegeek.github.io/e3nn_tutorial/ · NequIP <https://github.com/mir-group/nequip> · <https://www.nature.com/articles/s41467-022-29939-5> · MACE <https://github.com/ACEsuit/mace> · <https://arxiv.org/abs/2206.07697> · Allegro <https://github.com/mir-group/allegro> · Equiformer <https://arxiv.org/abs/2206.11990> · curated list <https://github.com/Eipgen/Neural-Network-Models-for-Chemistry> · AlphaFold 2 <https://www.nature.com/articles/s41586-021-03819-2> · ESM <https://github.com/facebookresearch/esm> · ESM Atlas <https://esmatlas.com/> · Boltz <https://github.com/jwohlwend/boltz> · OpenFold <https://github.com/aqlaboratory/openfold> · QM9 <https://quantum-machine.org/datasets/> · PCQM4Mv2 <https://ogb.stanford.edu/kddcup2021/pcqm4m/> · MoleculeNet <https://moleculenet.org/> · OGB <https://ogb.stanford.edu/>

Interview (P9) — NeetCode <https://neetcode.io/> · LeetCode <https://leetcode.com/> · DesignGurus <https://www.designgurus.io/> · Striver <https://takeuforward.org/> · ByteByteGo <https://bytebytego.com/> · <https://www.youtube.com/@ByteByteGo> · Gaurav Sen <https://www.youtube.com/@gkcs> · MIT 6.824 <https://pdos.csail.mit.edu/6.824/>

Company ladder — Qure.ai <https://qure.ai/> · SigTuple <https://sigtuple.com/> · NVIDIA <https://www.nvidia.com/en-us/about-nvidia/careers/> · GE Healthcare <https://www.gehealthcare.com/> · MSR India <https://www.microsoft.com/en-us/research/lab/microsoft-research-india/> · Google Research India <https://research.google/locations/india/> · Owkin <https://www.owkin.com/> · Recursion <https://www.recursion.com/careers> · Insitro <https://insitro.com/> · Genesis <https://www.genesis therapeutics.ai/> · Iambic <https://www.iambic.ai/> · Chai <https://www.chaidiscovery.com/> · EvolutionaryScale <https://www.evolutionaryscale.ai/> · Profluent <https://www.profluent.bio/> · InstaDeep <https://www.instadeep.com/> · Valence Labs <https://www.valencelabs.com/> · Isomorphic Labs <https://www.isomorphiclabs.com/careers> · Google DeepMind <https://deepmind.google/about/careers/> · QpiAI <https://www.qpi.ai/tech/careers> · IBM Quantum <https://www.ibm.com/quantum> · IISc ERP <https://www.iisc.ac.in/admissions/external-registration-programme-phd/> · EURAXESS <https://euraxess.ec.europa.eu/>

A note on what you're taking on

This is the definitive plan: ~2,530 hours over ~16.5 months, sequenced so nothing is left shaky — full math and fundamentals depth (intuition-first, generously timed), the complete imaging-through-video and CV-expert breadth, a protected geometric/molecular spike, a genuine quantum/QML build run after you're already interviewing, and threads sized for real fluency in coding, from-scratch implementation, and research discussion. The calendar is committed — Block 1 by ~April 2027, Block 2 by ~July–August, Phase Q and everything by end-2027 — but the **self-tests and Feynman gates are the true gates**: if a gate isn't passed, you let the block stretch rather than carrying a weak foundation forward. Lean on the consolidation weeks; they're what keep confusion from compounding over a long, aggressive push. Apply from ~month 2 on your existing applied skill, and let the incremental ladder carry you upward as each capstone lands.

On the ambition — landing among the very best labs in the world: no plan can *guarantee* a specific offer, because timing, headcount, visas, and the day itself all play a part. What this plan *can* do, and does, is make you a genuinely strong, well-rounded, honestly-positioned candidate for that tier — deep where it counts, broad enough to be unshakeable in an interview, with a differentiated spike (medical imaging + geometric/molecular ML bridged from quantum chemistry) that very few people can claim, and a body of public work to prove it. Bring the aggression you described to the *consistency* — the daily threads, the weekly Feynman gate, the monthly reproduced paper — more than to any single heroic week, and by end-2027 you'll walk into those rooms with the one thing the pain point was really about: the quiet confidence of someone who doesn't just use these ideas but can teach them. Adjust as you go; this is a scaffold for durable mastery and a real shot at the top.